Table 4-2. *Logical Operators*

Symbol	Description	Is true if…
&&	logical and	both expressions are true
||	logical or	either expression is true
!	logical not	the expression is false

Control Statements

Up until this chapter, our source code has been relatively straightforward. The Arduino microcontroller reads the lines of code in our sketches from the top of the code until it reaches the end at the bottom, and, when there are no more commands to execute, will bounce back to the top of the `loop()` function and start over again. The order or direction in which code is executed is called **program flow** and the ability to alter this order of executing commands is something called **flow control**.

There are two kinds of statements available in Arduino C for controlling program flow: **conditional** statements that make decisions based on whether or not a condition is true and **iterative** statements that perform things a certain number of times or until a condition becomes false. Conditional statements that include the `if` and `switch` statements selectively perform certain actions based on the state of a condition. Iterative statements that include the `for` and `while` statements are often called loops because, just like our `loop()` function, they will loop to the beginning of the statement and repeat their code when a condition has been met.

If

The `if` statement is the simplest of the control structures and is among the most prominent method for Arduino decision making. It will perform a block of code if a specific condition is met. The basic syntax for an `if` statement looks as follows:

```
if (condition) {
  statements
}
```

Following the keyword `if`, a pair of parentheses encloses a conditional expression. If we wanted to test a certain variable to see if it is equal to the value 5, we would begin the statement like the following:

```
if (myValue == 5)
```

In this example, if `myValue` is equal to 5, the statement would execute any following simple statement or block of statements enclosed by curly braces. If the condition is not met, in that they return the value false, then the following statement(s) will be ignored and are not executed.

A common use for the `if` statement is to read the state of a digital pin and perform an action based on its condition. As a hypothetical example, this could look something like the following:

```
tiltSwitch = digitalRead(switchPin);
if (tiltSwitch == HIGH) digitalWrite(13, HIGH);
```